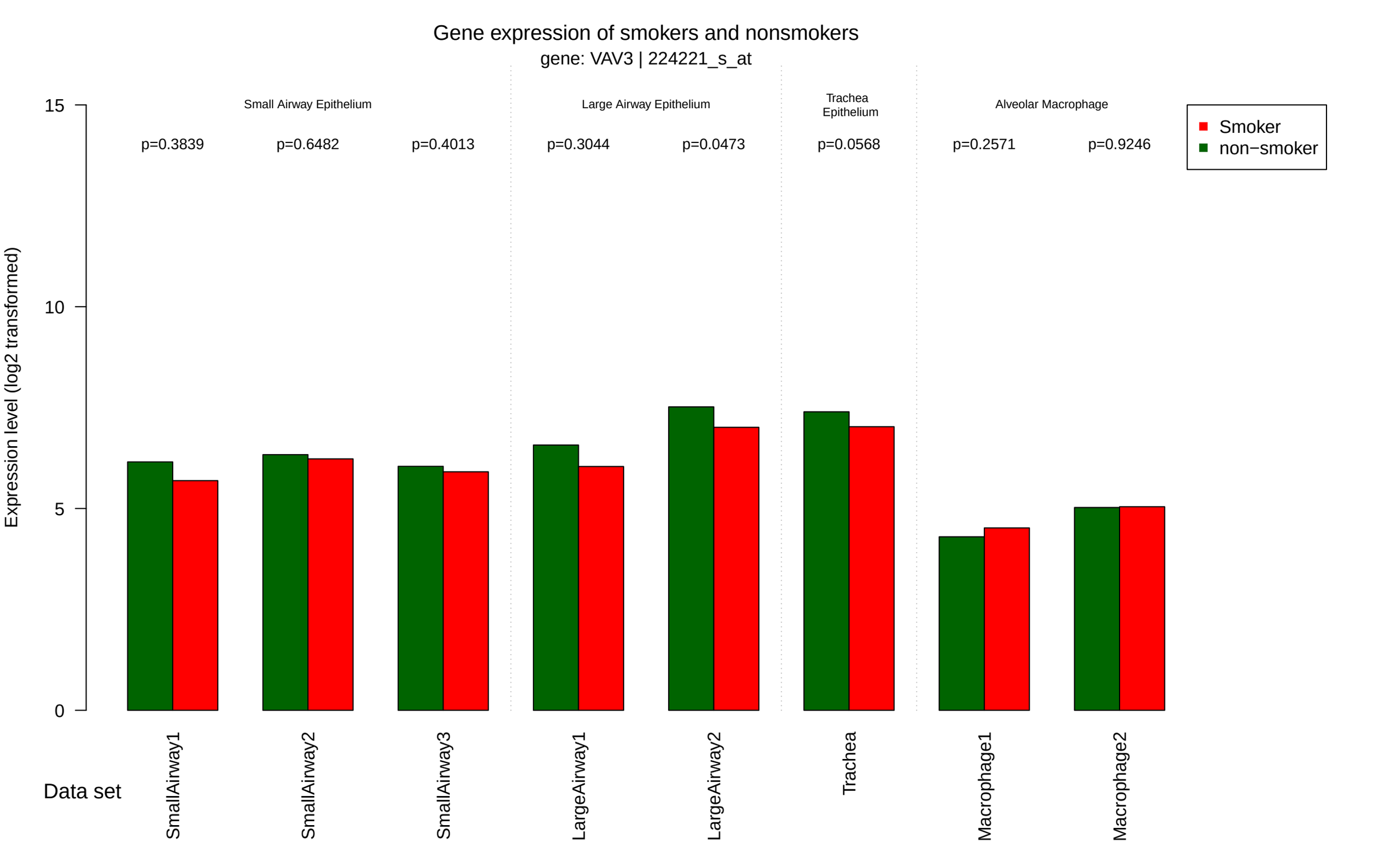
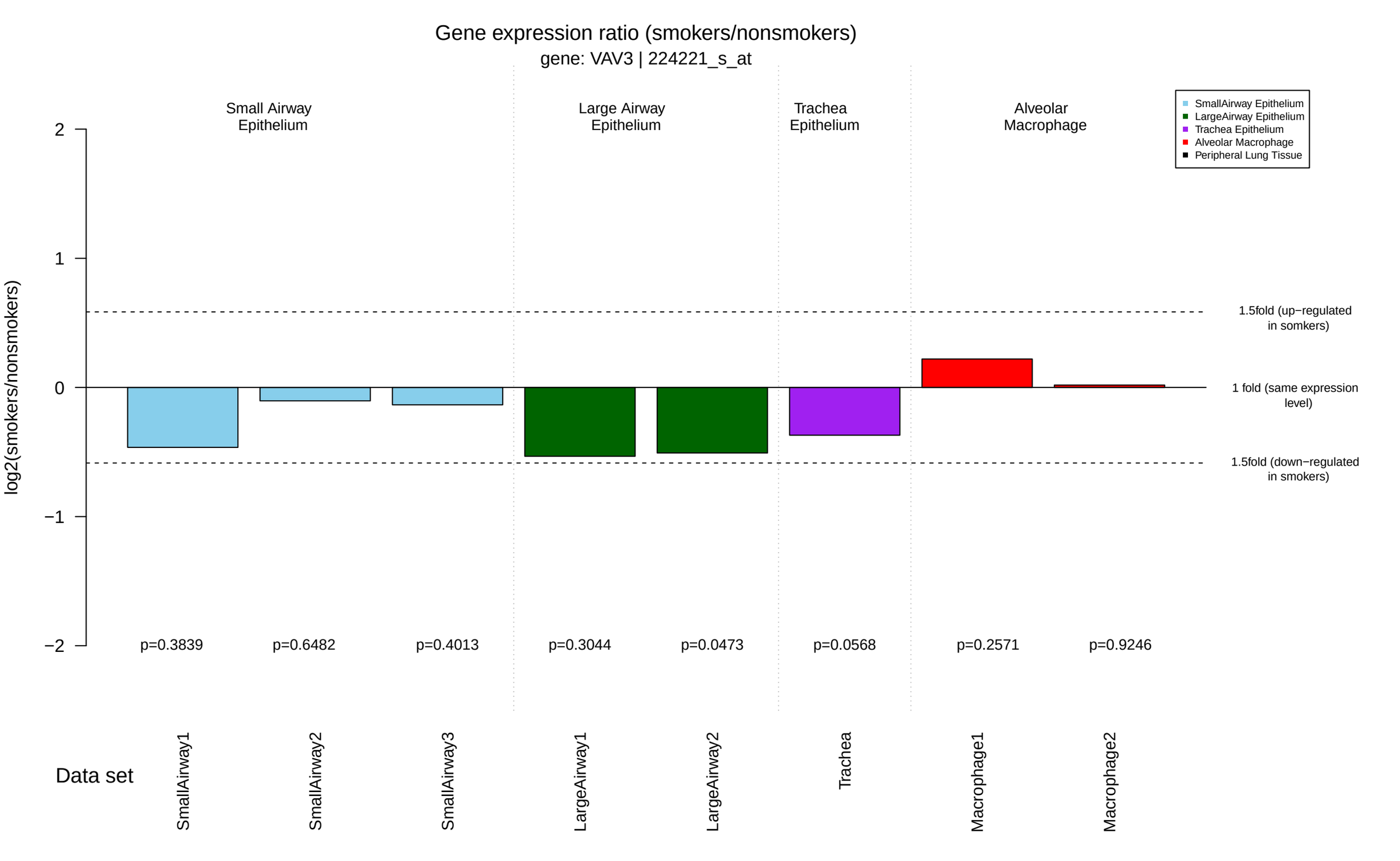
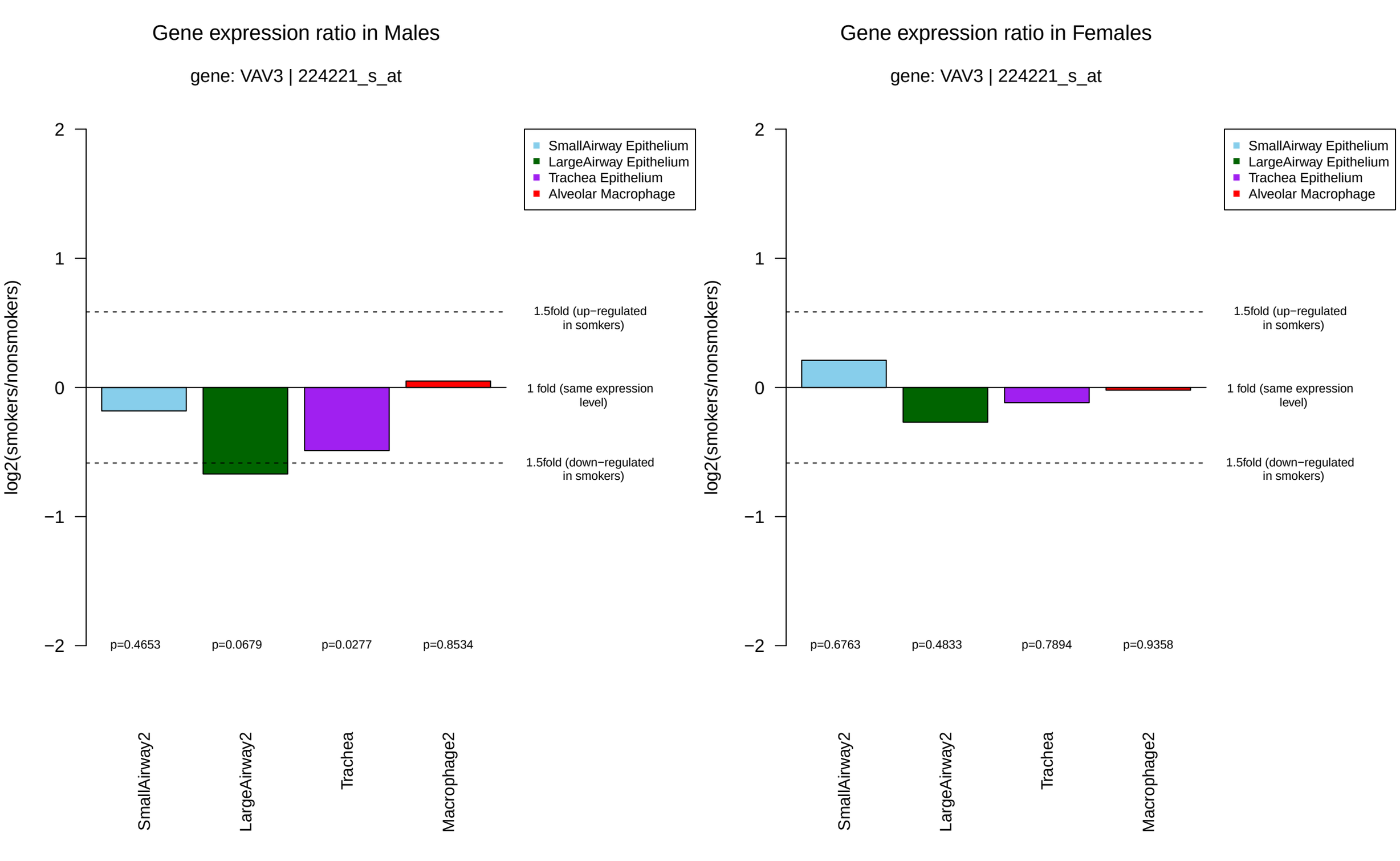
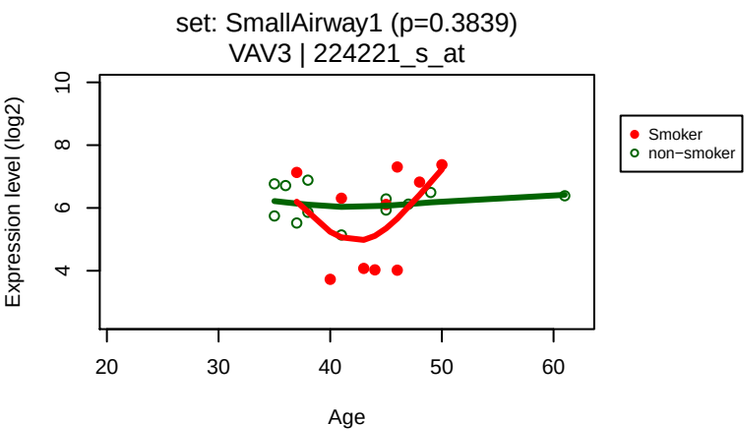
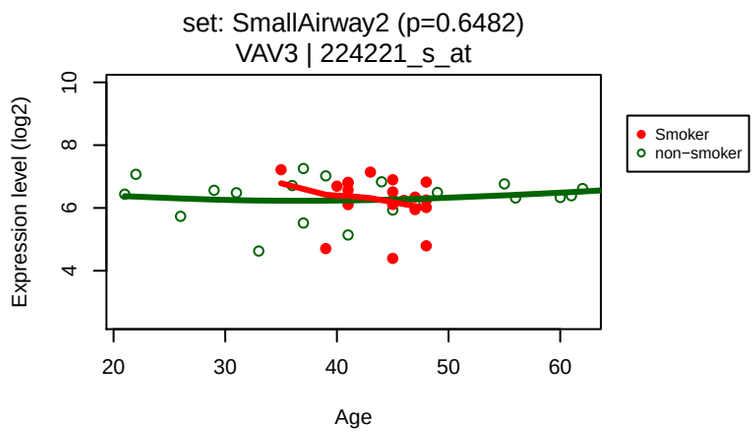
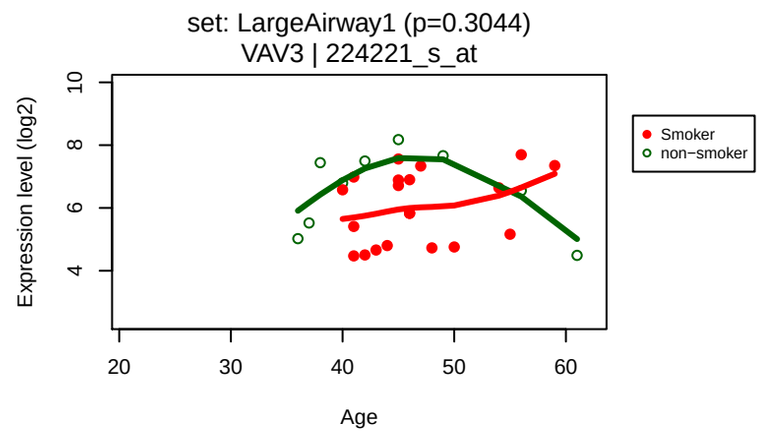

Gene expression of smokers and nonsmokers
gene: VAV3 | 224221_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3839
p=0.6482
p=0.4013
p=0.3044
p=0.0473
p=0.0568
p=0.2571
p=0.9246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

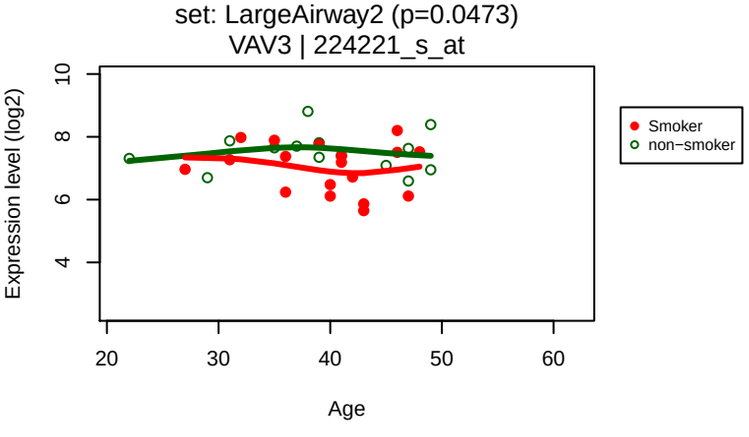

Gene expression ratio (smokers/nonsmokers)
gene: VAV3 | 224221_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3839
p=0.6482
p=0.4013
p=0.3044
p=0.0473
p=0.0568
p=0.2571
p=0.9246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

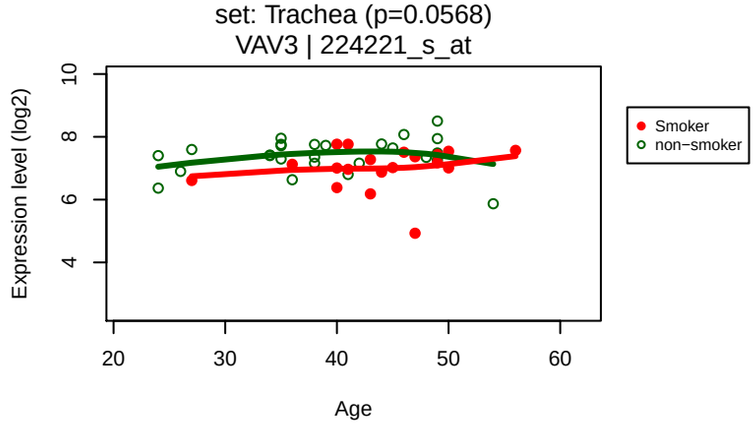

Gene expression ratio in Males
gene: VAV3 | 224221_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4653
p=0.0679
p=0.0277
p=0.8534
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VAV3 | 224221_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6763
p=0.4833
p=0.7894
p=0.9358
SmallAirway2
LargeAirway2
Trachea
Macrophage2

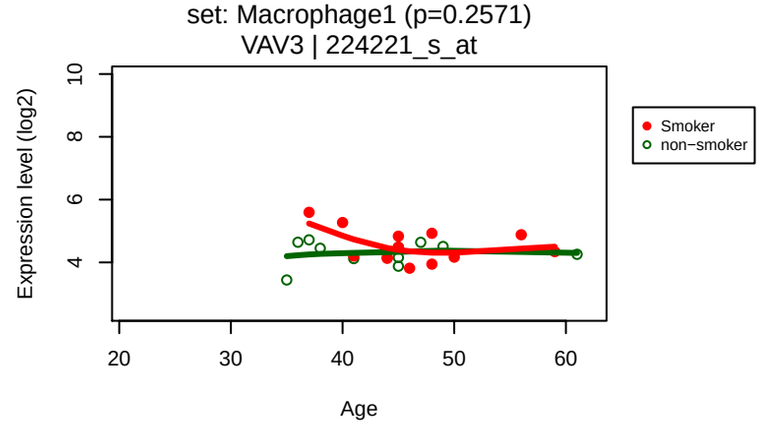

set: SmallAirway1 (p=0.3839)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker

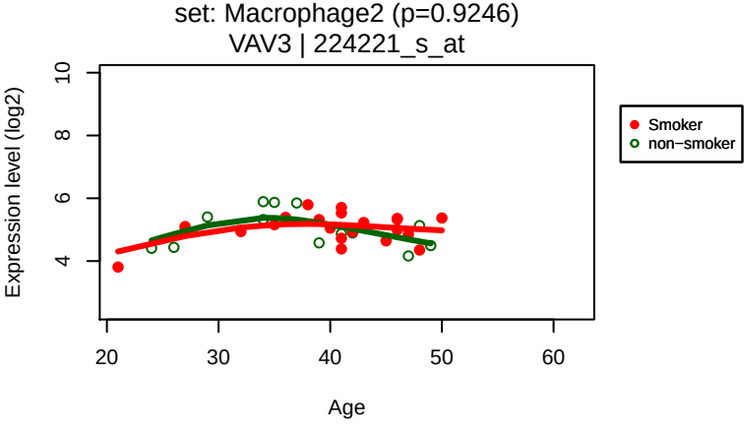

set: SmallAirway2 (p=0.6482)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker

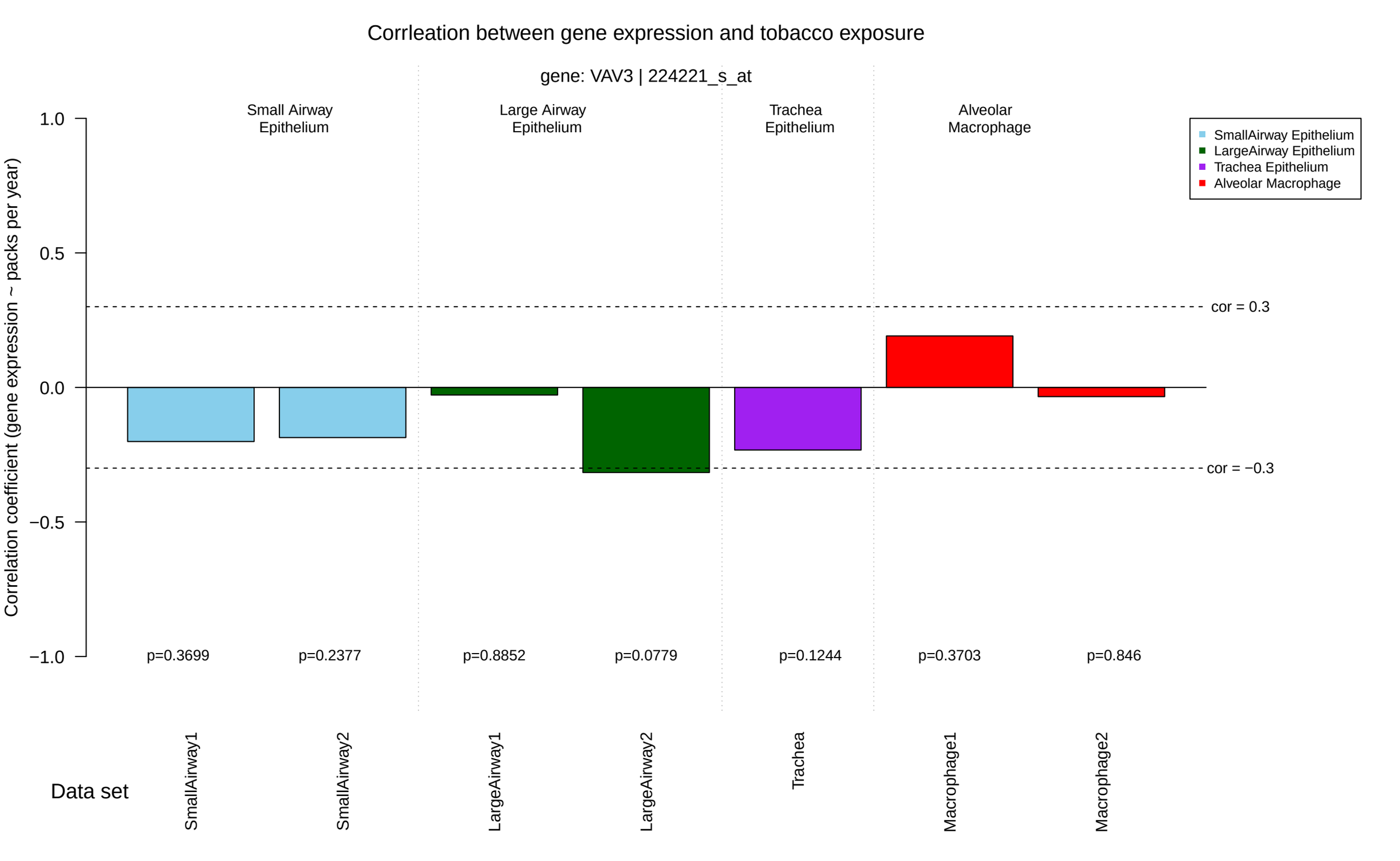

set: LargeAirway1 (p=0.3044)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0473)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0568)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2571)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9246)
VAV3 | 224221_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VAV3 | 224221_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3699
p=0.2377
p=0.8852
p=0.0779
p=0.1244
p=0.3703
p=0.846
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2